

Conor Burns

R&D Engineer

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Summary

Analytical R&D engineer with 2+ years of experience aiding product development. Specialties include: fixture creation/CAD, product testing, TMV, data analysis

Skills

Certifications: Certified SolidWorks Professional (CSWP)

Software: SolidWorks, Minitab, MATLAB, Excel, Bluehill, Copilot

Equipment: Instron, Zwick, Extensometer, Gurley, HVLD, HeLD, Lyophilizer, Bioprinter, ELISA

Work Experience

Performance Evaluation Engineer

Raritan, NJ

Johnson & Johnson MedTech

Sept 2025 – Present

- Designed a custom fixture to enable ISO-standard sealant travel distance validation at a horizontal expression angle, developing a dual rack-and-pinion mechanism to translate vertical Instron force into controlled horizontal motion; used fixture to differentiate high/low functionality groups using test data and authored a technical memo
- Analyzed product measurement data using ANOVA and percent difference in Minitab to evaluate measurement reduction opportunities; authored and presented a technical memo supporting a reduction in data point collection by 45%, facilitating faster data collection in subsequent studies
- Led as a subject matter expert of modified ASTM burst testing on 2 products as an expert operator, designing supplementary fixtures and performing Tukey, ANOVA, and T-tests in Minitab to evaluate product performance over 10+ studies and guide DOE and cross-functional development decisions
- Developed Bluehill test methods for tensile testing of implant materials using Instron, owning setup, calibration, and tests, ensuring accurate data capture for performance evaluation and downstream development decisions
- Fabricated an electromechanical test using a linear actuator and custom parts to automate device evaluation, enabling repeatable testing while eliminating the need for a \$3,000 commercial option
- Supported multiple R&D projects by developing tests, fabricating fixtures with SolidWorks, analyzing experimental data in Minitab, and generating visualizations to identify performance trends and inform engineering decisions

Research and Development Engineer Co-op

Edison, NJ

MTF Biologics

Jan 2024 – June 2024

- Simulated processing of biologic to decrease production time while maintaining critical product standards by reducing lyophilizer processing time by 33% through parameter optimization
- Reengineered an existing biologic using a novel production method, upcycling 8% of unutilized materials, and authoring an SOP for future replication

Undergraduate Research Assistant

Piscataway, NJ

Rutgers Musculoskeletal Regeneration Laboratory

May 2023 – Dec 2023

- Developed and implemented a bulk processing protocol, increasing process yield and reducing creation time by 20%
- Optimized multi-variable print parameters over 5+ trials to reliably extrude bioresin materials via 3D bioprinting
- Coordinated multi-week schedules for a team of 4, completing projects on time and improving turnaround by 20%

Education

Rutgers University – School of Engineering

New Brunswick, NJ

BS in Biomedical Engineering

Sept 2021 – May 2025